

DUCKDB 101

FROM DESKTOP TO NOTEBOOK

Fast, local analytics with SQL — then take it into notebooks
and data workflows.

NO SERVER. NO SETUP. JUST SQL.



Day of Data 2026 | Albany

Jason Romans | The DAX Shepherd

1 Desktop
start local

2 SQL
query files

3 Notebook
repeatable workflow

4 Fabric
cloud notebook
workflow

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HI, I'M JASON

THE DAX SHEPHERD*

*More semantic models, fewer sitcoms.



Honored to be
recognized by
Microsoft.

I HERD THE DATA



Microsoft MVP



Nashville, Tennessee



Fabric + Power BI



Teaching + Speaking



MY FAVORITE MODEL

TARDIS

TOOLS & ALLIES



Power BI



Microsoft Fabric



DuckDB



Python



SQL



VS Code

TAKE THIS WITH YOU

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All slides, resources, and demo files
available:

THE DAX SHEPHERD

thedaxshepherd.com/presentations



The DAX Shepherd

thedaxshepherd.com/presentations

Standing on the Shoulders of Giants

Grateful for the people who share their knowledge, encourage us, answer questions, and help the rest of us learn faster.



Photo credit: Unknown Author, CC BY-ND

SESSION ROADMAP

1. Intro

What DuckDB is and why it matters



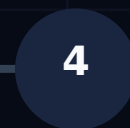
Intro



Desktop



Notebook



Fabric



Wrap-Up

NOW 1. Intro

What DuckDB is and why it matters

Next: Desktop

SESSION ROADMAP

2. Desktop

Running DuckDB locally — CLI and GUI



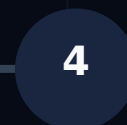
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Wrap-Up

NOW 2. Desktop

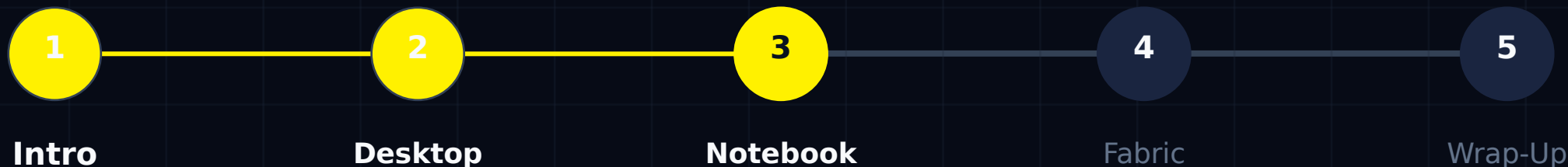
Running DuckDB locally — CLI and GUI

Next: Notebook

SESSION ROADMAP

3. Notebook

Interactive querying in VS Code



NOW 3. Notebook

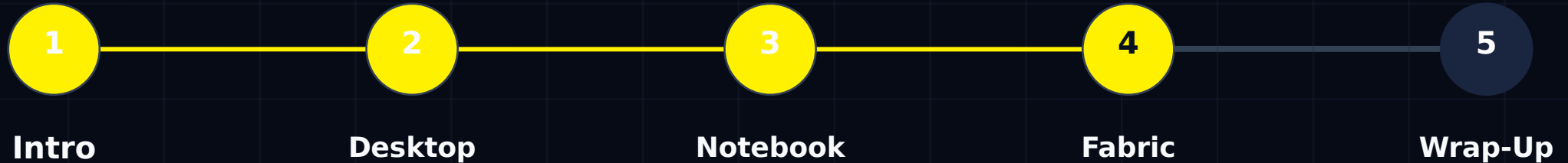
Interactive querying in VS Code

Next: Fabric

SESSION ROADMAP

4. Fabric

Fabric Notebook — plus a note on Power BI



NOW 4. Fabric

Fabric Notebook — plus a note on Power BI

Next: Wrap-Up

SESSION ROADMAP

5. Wrap-Up

Takeaways and what's next



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Wrap-Up

NOW 5. Wrap-Up

Takeaways and what's next

Thanks!

SESSION ROADMAP

1. Intro

What DuckDB is and why it matters



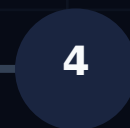
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Wrap-Up

NOW 1. Intro

What DuckDB is and why it matters

Next: Desktop

What is DuckDB?

The first mental model: an analytical database that runs inside your process.

**SQLite is embedded for apps.
DuckDB is embedded for
analytics.**

In-process

Runs in your app, Python process, notebook, or command-line session. No separate database server is required for the core workflow.

Analytical SQL

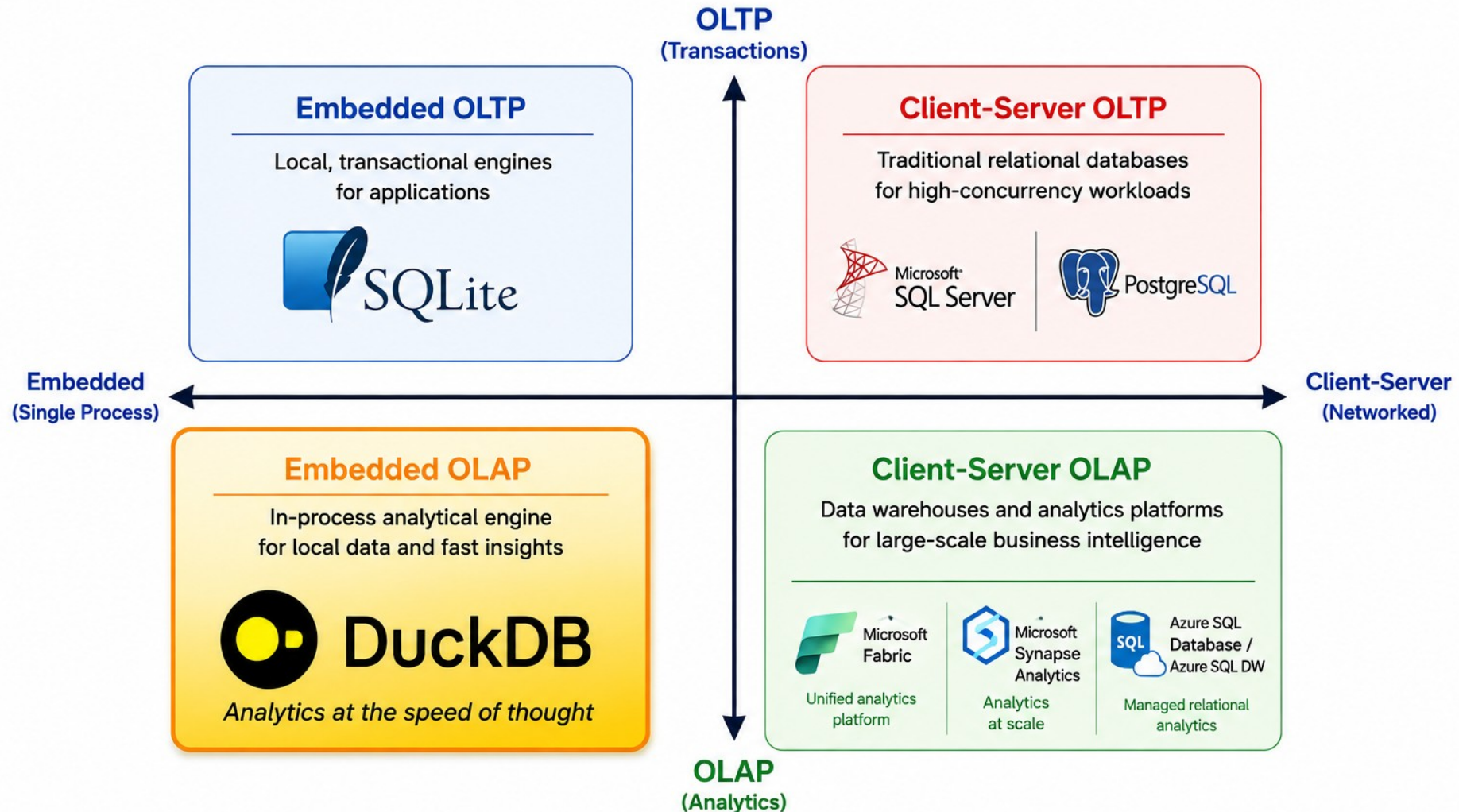
Optimized for OLAP-style questions: filtering, grouping, joining, scanning columns, and reading data files.

File-native

Can query files like CSV and Parquet directly, which makes it great for exploration and lightweight ETL.

The Data Systems Landscape

Where DuckDB Fits



Examples shown are representative, not exhaustive.

The Problem DuckDB Solves

Before DuckDB, running SQL on files meant spinning up infrastructure. Here's what changed.

Before DuckDB

- Need a server to run SQL
- Pandas loads the full dataset into memory
- Polar – not SQL
- Spark requires a cluster
- SQLite is row-oriented, not OLAP
- Moving data just to query it is slow

With DuckDB

- Run SQL on CSV, Parquet, JSON in place
- Out-of-core execution for larger-than-RAM data
- Often faster than Spark on one machine
- One pip install
- Query files where they live: local, S3, OneLake

Same SQL you already know

- Window functions, CTEs, PIVOT, UNPIVOT, ASOF joins
- No new query language to learn
- Works in CLI, Python, Jupyter, and Fabric notebooks

Use DuckDB before the big engine, not instead of thinking.

Who is DuckDB for?

A local analytics engine for people who need fast answers without a platform ceremony.



Data analysts exploring files locally

CSV, Parquet, Excel exports, one-off downloads, vendor extracts.



Engineers prototyping transformations

Test ETL logic locally before deciding whether it belongs in Spark, dbt, a warehouse, or Fabric.



Python users who still want SQL

Stay in SQL instead of fighting pandas or polars syntax — drop into a DataFrame only when you need to.



Speakers, teachers, and demo builders

Small setup surface area. Great for repeatable demos and hands-on sessions.

**Not just a database.
A shortcut to answers.**

Who is it not for?

DuckDB is powerful, but the best tool is still situational.

Not your high-concurrency OLTP app

If you need many users writing transactions all day, reach for Postgres, SQL Server, or another operational database.

Not a replacement for your warehouse

Central security, governance, shared semantic models, concurrency, and BI serving still matter.

Not a magic replacement for Spark

DuckDB is single-node; Spark is built for clusters. When data outgrows one machine, Spark is still the big engine.

Use DuckDB before the big engine, not instead of thinking.

What many people see

DuckDB



The deeper DuckDB ecosystem



*There's more
below the surface.*



SQL on files



CSV / Parquet / JSON



Local analytics



CLI



Python



VS Code + Notebooks



Extensions



MotherDuck



DuckLake



Data prep



Quack



UI



Power BI + Fabric workflows

Why DuckDB belongs in your toolkit

1

Local & free

Runs on your machine. No server, no install ceremony, no licensing. One binary, instant SQL.

SQL ✓

2

Reads files directly

Query CSV, Parquet, and JSON in place. The 671 MB flights dataset with 6.8M rows Excel can't open? DuckDB scans it in seconds.

SQL ✓

3

Scales with your workflow

Same SQL works in CLI, Python, Jupyter, and Fabric notebooks. Learn once, use everywhere.

SQL ✓

Why should a Power BI person care?

1

Inspect before import

Look at large CSV and Parquet files before bringing them into Power BI.

Profile files that Excel or Power Query may struggle with.

2

Prototype locally

Test SQL transformation logic on your laptop before you commit to a model, pipeline, or warehouse pattern.

Convert raw files into cleaner Parquet.

3

Bridge to Fabric

Use VS Code notebooks for repeatable local analysis.

Carry the same exploration pattern forward into Fabric workflows.

DuckDB vs. Power Query

Different tools for different moments in the data workflow.

Use DuckDB when...

- Data is too large for Power Query
- You want SQL over CSV or Parquet
- Profiling files before import
- Prototyping ETL logic locally
- Exporting clean Parquet for downstream

Use Power Query when...

- Loading data into a Power BI model
- M transforms and Power BI dataflows
- Governed refresh in Power BI Service
- Building reports without code
- Team-managed semantic models

DuckDB is for exploration and prep. Power Query is for the model.

Where DuckDB fits for Power BI people

Power BI + Fabric lens

Use it before Power BI/Fabric to inspect, shape, and hand off files — not as another platform to manage.



The practical pattern

DuckDB is the small engine you reach for when files are messy, too big for comfort, or need a quick repeatable check.

1

Before import

Profile large CSV/Parquet files before they become a Power Query problem.

2

During exploration

Use SQL + Python in a local VS Code notebook; uv keeps it repeatable.

3

Before handoff

Export clean Parquet or validated logic, then hand off to Power BI or Fabric.

Demo path: Local DuckDB → Files as Tables → VS Code Notebook → Power BI + Fabric

Where would you use DuckDB?

A lightweight analytics engine for local exploration, files, notebooks, and the Microsoft data stack.

Local DuckDB

Start with a local engine

Open DuckDB from the CLI, an in-memory database, or a small .duckdb file. No server or cluster needed.

Files as Tables

Query data where it already lives

Use SQL directly against CSV, Parquet, JSON, and folders before deciding what to import or convert.

VS Code Notebook

Make exploration repeatable

Use Jupyter in VS Code with uv so SQL, Python, charts, and documentation live in the same local workflow.

Power BI + Fabric

Bridge back to the Microsoft stack

Run DuckDB in Fabric Python notebooks: read Lakehouse files, do light SQL transforms, skip Spark when you don't need it.

THE MENTAL MODEL

Start local, query files, then move the pattern into notebooks and Fabric.

SESSION ROADMAP

2. Desktop

Running DuckDB locally — CLI and GUI



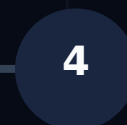
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Wrap-Up

NOW 2. Desktop

Running DuckDB locally — CLI and GUI

Next: Notebook

Install, launch, and run your first query

WINDOWS • MAC • LINUX

```
# Windows
winget install DuckDB.cli

# Mac
brew install duckdb
```

LAUNCH

```
duckdb
```

FIRST QUERY

```
SELECT 'DuckDB is running' AS demo;
-- Don't forget the semicolon!
```

Don't forget the semicolon!

WATCH OUT

Exit thrashing

``quit`` and ``exit`` don't work. DuckDB uses dot-commands for shell control:

```
.exit or .quit
.help
.databases
.tables
```

Query files directly — no loading required



flights_2023.csv — 671 MB, 6.8M rows

Excel can't open it. DuckDB scans it in seconds — straight from the file.



SELECT FROM A CSV

```
SELECT * FROM 'flights_2023.csv' LIMIT 10;
```

No CREATE TABLE. No import. Just point at the file.

DESCRIBE THE SCHEMA

```
DESCRIBE SELECT * FROM 'flights_2023.csv';
```

Inspect columns and types without loading data.

SUMMARIZE EVERYTHING

```
SUMMARIZE SELECT * FROM 'flights_2023.csv';
```

Stats for every column in one command — DuckDB's secret weapon.

Same SQL, three interfaces

CLI for speed → Python script for repeatability → Notebook for exploration. One uv environment owns it all.

STEP 1

CLI

Pure SQL.
Fastest path to first query.

```
duckdb
SELECT * FROM
'flights_2023.csv';
```



STEP 2

Python (uv)

Project owns its environment.
No pip drift.

```
uv init
uv add duckdb pandas
uv run python main.py
```



STEP 3

Notebook

Same kernel, interactive.
VS Code or Jupyter.

```
uv add jupyter ipykernel
code .
```

The notebook isn't a different system — it's another interface to the same environment.

DuckDB Extensions

Install with `INSTALL extension_name;` `LOAD extension_name;` — or let DuckDB auto-load on first use.

httpfs

Query files over HTTP/S and S3 — no download step. Essential for remote data.

azure

Azure Blob Storage, ADLS, and OneLake — query Fabric files with the same SQL.

excel

Read .xlsx files directly. `SELECT * FROM 'my_file.xlsx'` just works — no conversion needed.

delta / iceberg

Native Delta Lake and Apache Iceberg table support. Read lakehouse tables without Spark.

spatial

Geometry types, spatial joins, and PostGIS-compatible functions for geospatial analysis.

DuckDB's Built-in UI

duckdb --ui — opens at localhost:4213. Runs fully local; your data never leaves the machine.

The screenshot displays the DuckDB built-in UI interface. The top bar includes the DuckDB logo, a search bar, and links for "Sign in to MotherDuck" and "Help".

Left Sidebar:

- Search
- Notebooks
 - My Notebook
 - DuckDB UI basics
- Attached databases
 - memory
 - main
 - trains

Query Editor:

```
1  /**
2   Exploring Column Diagnostics
3
4   When running a query, your results will appear below the editor
5   and column diagnostics will appear on the right panel, providing an
6   overview of your data without requiring additional queries.
7
8   Run this query, then click on column names in the right panel to view detailed statistics.
9   */
10
11  FROM trains;
```

Results: First 50,000 rows returned in 65ms

	123 service_id	date	T type	123 train_number	T station_code	T station_name	
5	11196117	2023-05-15	Intercity	1410	SHL	Schiphol Airport	26
6	11196117	2023-05-15	Intercity	1410	ASD	Amsterdam Centraal	26
7	11196117	2023-05-15	Intercity	1410	UT	Utrecht Centraal	NI
8	11196118	2023-05-15	Nightjet	420	NURNB	Nürnberg Hbf	26
9	11196118	2023-05-15	Nightjet	420	FFS	Frankfurt (Main) Süd	26
10	11196118	2023-05-15	Nightjet	420	FFFM	Frankfurt (M) Hbf	NI
11	11196118	2023-05-15	Nightjet	420	FNAF	Frankfurt Flughafen Fernb	26
12	11196118	2023-05-15	Nightjet	420	FMZ	Mainz Hbf	26
13	11196118	2023-05-15	Nightjet	420	KKO	Koblenz Hbf	26
14	11196118	2023-05-15	Nightjet	420	BONN	Bonn Hbf	26
15	11196118	2023-05-15	Nightjet	420	KKW	Köln West	NI
16	11196118	2023-05-15	Nightjet	420	KN	Kaldenkirchen	NI
17	11196118	2023-05-15	Nightjet	420	VL	Venlo	NI

Column Diagnostics:

380,959 Rows
8 Columns

Column	Statistics
123 service_id	16
123 train_number	541
T station_code	598
T station_name	11%
departure_time	11%
arrival_time	11%

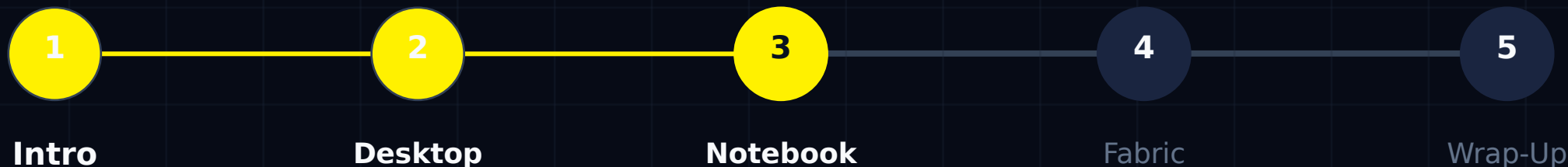
DEMO TIME

Desktop

SESSION ROADMAP

3. Notebook

Interactive querying in VS Code



NOW 3. Notebook

Interactive querying in VS Code

Next: Fabric

Why notebooks?

A notebook is not a new system. It is another interface to the same environment.

Scripts are great for

- Repeatable, automated execution
- Clean, version-controlled logic
- Production pipelines and scheduled jobs
- Reproducible runs from a known state

Notebooks are great for

- Step-by-step exploration of files and data
- Teaching, demos, and live walkthroughs
- Visible results next to the SQL that made them
- Quick iteration when shape and intent are still evolving

Same environment. Same DuckDB. Different interaction model.

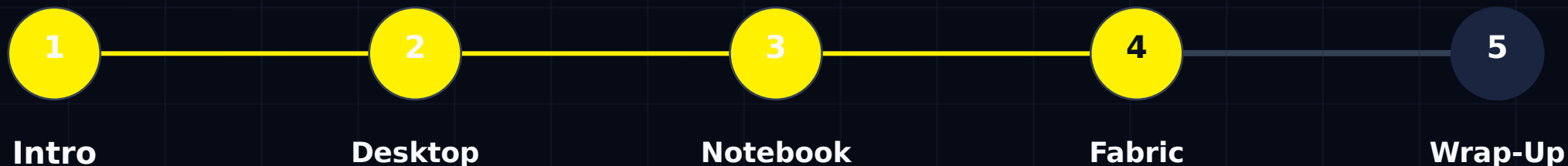
DEMO TIME

Notebook

SESSION ROADMAP

4. Fabric

Fabric Notebook — plus a note on Power BI



NOW 4. Fabric

Fabric Notebook — plus a note on Power BI

Next: Wrap-Up

Power BI + DuckDB

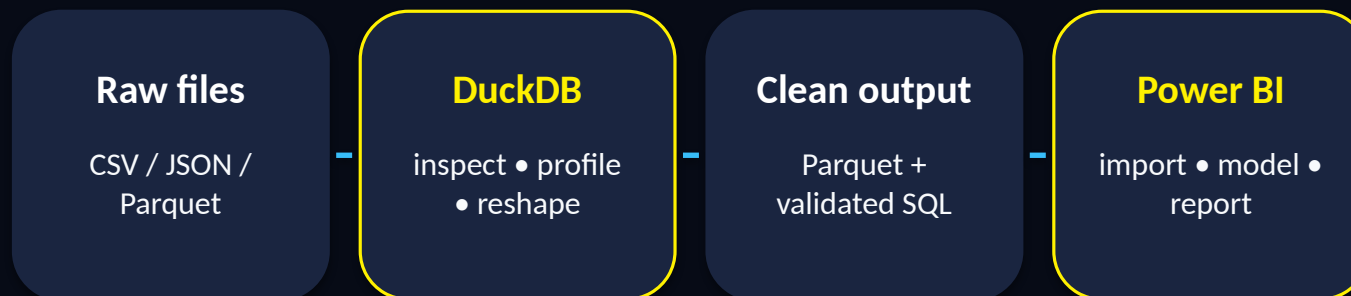
Use DuckDB before the model — not necessarily as the model's live source.

Power BI pattern

DuckDB is usually most valuable as a prep, profiling, and validation tool before Power BI imports the data.

Inspect large CSV / Parquet files
Prototype SQL transformations locally
Export cleaner, typed Parquet

Practical message: DuckDB before Power BI.



Connection options exist, but they are not the main 101 path

Possible via ODBC + MotherDuck Power Query connector — works, but not the main 101 path.

Same notebook, different runtime

When you move the notebook into Microsoft Fabric, the SQL stays. Three things change.

✓ STAYS THE SAME

- DuckDB SQL syntax
- Python code
- Query patterns
- Notebook structure
- Your mental model

→ WHAT CHANGES

Environment

`uv (you own it)` → Fabric runtime

Data paths

`../data/flights/*.csv` →
`/lakehouse/default/Files/`

Install

`uv add duckdb` → pre-installed in Fabric

DECISION POINT

Reach for DuckDB when:

Quick SQL exploration · Small to medium data · Inspecting files ·
Avoiding Spark startup

Reach for Spark when:

Data is large · Workloads need to scale · Production pipelines ·
Distributed processing

Fabric + DuckDB

Use the smallest engine that can safely do the job.

In Fabric, DuckDB fits naturally in a Pure Python notebook when you want quick single-node SQL over files.

Pure Python notebook + DuckDB

Good when the task is local-ish and exploratory:

Inspect files in OneLake / Lakehouse Files

Run quick SQL over CSV or Parquet

Prototype transformation logic

Avoid starting Spark for small-to-medium work

Capacity angle: lower-overhead exploration.

Spark notebook

Use when the workload needs distributed scale:

Very large data or heavy joins

Production pipelines

Spark ecosystem features

Distributed processing across the cluster

Decision point: do I actually need Spark?

Same idea, new path: `read_parquet('/lakehouse/default/Files/flights_2023.parquet')`

DuckDB vs Spark in Fabric

Choose the engine based on the workload, not the logo.

Use DuckDB when...

- Quick SQL exploration
- Small-to-medium files
- Notebook-first investigation
- Avoiding Spark startup for tiny jobs
- File inspection and prototypes

Use Spark when...

- Large/distributed workloads
- Production pipelines
- Lakehouse table processing
- Scale-out transformations
- Operationalized data engineering

DuckDB is the pocketknife. Spark is the workshop.

DEMO TIME

Fabric

SESSION ROADMAP

5. Wrap-Up

Takeaways and what's next



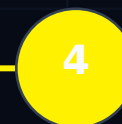
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Wrap-Up

NOW 5. Wrap-Up

Takeaways and what's next

Thanks!

A practical use-case ladder

DuckDB often starts as exploration, then becomes a repeatable step.

Explore

What is in this file?

`read_csv_auto / read_parquet`

Profile

What are the shapes, nulls, and weird values?

`COUNT, GROUP BY, DESCRIBE`

Transform

Can I clean or reshape this quickly?

`CREATE TABLE AS SELECT`

Export

Can I hand off a better format?

`COPY TO Parquet`

Prototype

Does this ETL logic work before production?

`Python + SQL + tests`

Official DuckDB Resources

Start here: official docs, official sites, and free learning offers.

OFFICIAL

DuckDB

Main project site, docs, installation, guides, SQL reference, and client APIs.

<https://duckdb.org>

OFFICIAL DOCS

DuckDB Documentation

The source of truth for installation, CLI, SQL, CSV/Parquet, Python, and extensions.

<https://duckdb.org/docs/current/>

CLOUD + LEARNING

MotherDuck

Cloud analytics built around DuckDB, plus practical articles, videos, and learning resources.

<https://motherduck.com>

FREE BOOK OFFER

DuckDB in Action

Free PDF from MotherDuck. Good practical next step after this 101 session.

motherduck.com/duckdb-book-brief

FREE EARLY ACCESS / PREVIEW

DuckLake: The Definitive Guide

O'Reilly early release sponsored by MotherDuck. Chapters are released as they become available.

motherduck.com/lp/ducklake-lakehouse-table-format-book-full

Keep the first resource slide boring on purpose: official docs, official project, and current free offers.

Go Beyond

Books, videos, and community resources when you want more than the docs.

Books

DuckDB: Up and Running

O'Reilly

Good for a structured walk-through of DuckDB concepts and hands-on usage.

DuckDB in Action

Manning / free PDF via MotherDuck

Practical book for learning DuckDB from ingestion through analytics workflows.

DuckLake: The Definitive Guide

O'Reilly Early Release

Preview / early access. Best after DuckDB basics when you want lakehouse table-format context.

Video + community

VIDEO

MotherDuck YouTube

Product walkthroughs, livestreams, DuckDB ecosystem videos, and DuckLake content.

<https://www.youtube.com/@MotherDuckDB>

LEARNING PATH

Luke Barousse

Beginner-friendly SQL learning using DuckDB and MotherDuck in practical examples.

<https://www.youtube.com/@datawithluke>

COMMUNITY

GitHub + DuckDB Community

Source code, issues, discussions, extension ecosystem, and community links.

<https://duckdb.org/community>

Suggested path: docs first → free DuckDB in Action PDF → videos/community → O'Reilly books as you go deeper.

THANK YOU



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Slides



Session Evaluation